

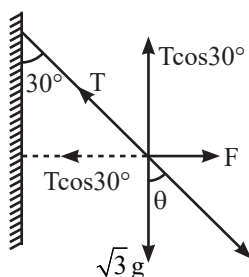


ANSWER KEY

1. (a) 2. (a) 3. (b) 4. (a) 5. (c) 6. (a) 7. (c) 8. (b) 9. (d) 10. (c)
 11. (c) 12. (b) 13. (c) 14. (b) 15. (d) 16. (b) 17. (b) 18. (b) 19. (b) 20. (c)
 21. (b) 22. (a) 23. (c) 24. (b) 25. [12] 26. [3] 27. [12] 28. [6] 29. [164] 30. [492]

Hint & Solutions

1. (a)



$$T \cos 30^\circ = mg$$

$$\cos 30^\circ = \frac{\sqrt{3}g}{T} \Rightarrow \frac{\sqrt{3}}{2} = \frac{\sqrt{3}g}{T}$$

$$\Rightarrow T = 20 \text{ N}$$

2. (a) $\vec{v} = 2t\hat{i} + 3t^2\hat{j}$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 6t\hat{j}$$

$$\text{When } t = 1 \text{ sec} \Rightarrow \vec{a} = 2\hat{i} + 6\hat{j}$$

$$\therefore \vec{F} = m\vec{a} = 0.5(2\hat{i} + 6\hat{j}) = \hat{i} + 3\hat{j}$$

$$\hat{i} + 3\hat{j} = i + 3\hat{j} \Rightarrow x = 3$$

3. (b) Fact based.

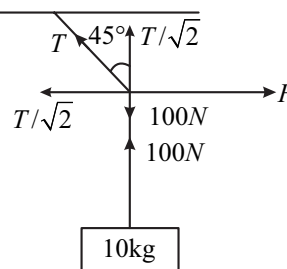
4. (a) Given, $\vec{a} = \frac{dv_x}{dt}\hat{x} + \frac{dv_y}{dt}\hat{y}$

$$\therefore \frac{dv_x}{dt} = \alpha \frac{dy}{dt} = 2\alpha^2 x$$

$$\frac{dv_y}{dt} = 2\alpha v_x = 2\alpha^2 y$$

$$\vec{F} = m\vec{a} = 2m\alpha^2(x\hat{x} + y\hat{y})$$

5. (c)



$$\frac{T}{\sqrt{2}} = 100$$

$$\frac{T}{\sqrt{2}} = F$$

$$F = 100 \text{ N}$$

$\Rightarrow$ The graph given in option (a) suits the best for the above relation.

6. (a) Let additional force is $\vec{F}$, so that $F_{\text{net}} = 0$

$$\vec{F} + 5(+\hat{i}) + 6(-\hat{i}) + 7(+\hat{j}) + 8(-\hat{j}) = \vec{0}$$

$$\Rightarrow \vec{F} - (\hat{i} + \hat{j}) = \vec{0} \Rightarrow \vec{F} = \hat{i} + \hat{j}$$

$$\Rightarrow |\vec{F}| = \sqrt{1^2 + 1^2} = \sqrt{2} \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{\text{Component of } \hat{j}}{\text{Component of } \hat{i}} \right)$$

$$\theta = \tan^{-1} \left(\frac{1}{1} \right) = 45^\circ$$

7. (c) $Ma = T - Mg$ (While climbing up)

$$T_1 = M(a + g)$$

$$T_1 = 50(5 + 10)$$

$$= 750 \text{ N}$$

(Rope will break)

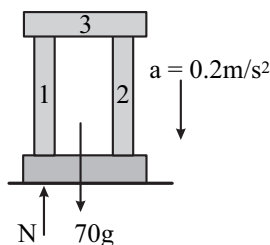
8. (b) From Newton's second law, $\vec{a} = \frac{\vec{F}}{m} = 8\hat{i} + 2\hat{j}$

$$\text{From Kinematic equation, } \vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$\vec{r} = 0 + \frac{1}{2}(8\hat{i} + 2\hat{j})10^2$$

$$\vec{r} = 400\hat{i} + 100\hat{j}$$

9. (d) Using free body diagram given below.

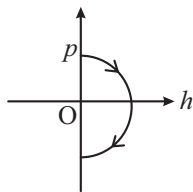


$$70g - N = 70a = 70 \times 0.2 = 14$$

$$\Rightarrow N = 700 - 14 = 686\text{N}$$

10. (c) Momentum $p = mv$

$$\text{Motion under gravity } h = \frac{u^2 - v^2}{2g}$$

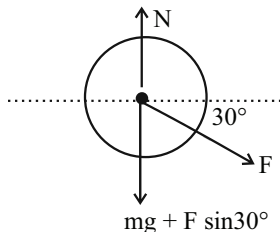


$$h = \frac{u^2 - p^2 / m}{2g} \quad (\because p = mv)$$

11. (c) Equating the vertical forces $N = mg + F \sin 30^\circ$

$$= 700 + 200 \times \frac{1}{2}$$

$$= 800\text{ N.}$$



12. (b) Impulse = Area under $F - t$ curve

$$(a) \text{ Impulse} = \frac{1}{2} \times 1 \times 0.5 = \frac{1}{4} \text{ N.s}$$

$$(b) \text{ Impulse} = 0.5 \times 2 = 1 \text{ N.s (maximum)}$$

$$(c) \text{ Impulse} = \frac{1}{2} \times 1 \times 0.75 = \frac{3}{8} \text{ N.s}$$

$$(d) \text{ Impulse} = \frac{1}{2} \times 2 \times 0.5 = \frac{1}{2} \text{ N.s}$$

$$13. (c) \left| \frac{d\vec{p}}{dt} \right| = |\vec{F}| \Rightarrow \frac{d\vec{p}}{dt} = \text{Slope of curve}$$

max slope (c), $\Rightarrow$ min slope (b)

$$14. (b) P_i = Nmv\hat{i} \text{ and } \vec{P}_f = -Nmv\hat{i}$$

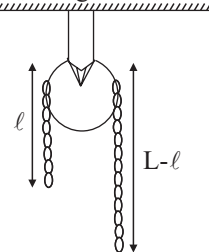
where N is the number of balls that strike the wall

$$\Delta \vec{P} = \vec{P}_f - \vec{P}_i = -2Nmv\hat{i} = -200 Nmv\hat{i}$$

$$\vec{F}_{\text{Total}} = \frac{\Delta \vec{P}}{\Delta t} = -\frac{200mvt}{t}$$

$$|\vec{F}| = \frac{200mv}{t}$$

15. (d) Let, Mass = m , Length = L ,



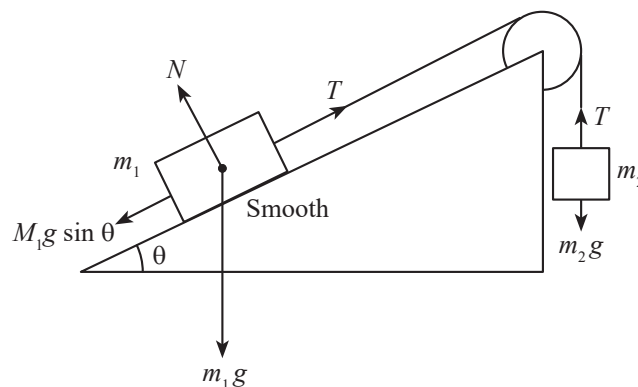
$$\frac{m}{L} = \lambda$$

$$a = \frac{F_{\text{Pulling}}}{\text{mass}} = \frac{(L-l)\lambda g - \lambda l g}{\lambda L} = \frac{(L-2l)\lambda g}{\lambda L} = \frac{g}{2}$$

$$\Rightarrow L - 2l = \frac{L}{2}; \quad l = \frac{L}{4} \Rightarrow x = 4$$

16. (b) As m_1 is in rest

$$T = m_2 g = 30 \text{ N}$$



$$T = m_1 g \sin \theta \Rightarrow 50 \sin \theta = 30$$

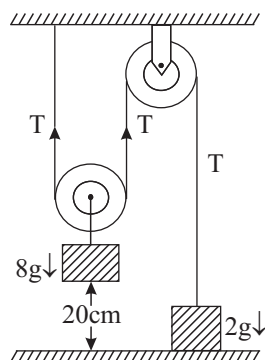
$$\Rightarrow \sin \theta = \frac{3}{5} \Rightarrow \theta = 37^\circ$$

$$\therefore N = 50 \cos \theta = 50 \cos 37^\circ = 50 \times \frac{4}{5} = 40N$$

$$17. (b) a_1 = \frac{2M_1g - M_1g}{2M_1 + M_1} = \frac{g}{3}$$

$$a_2 = \frac{3M_1g - M_1g}{3M_1 + M_1} = \frac{g}{2} \therefore \frac{a_1}{a_2} = \frac{2}{3}$$

18. (b) From the free body diagram,



For 8 kg block,

$$8g - 2T = 8a$$

...(i)

For 2 kg block,

$$T - 2g = 2 \times 2a$$

...(ii)

Solving equations (i) and (ii)

$$a = \frac{g}{4} = 2.5 \text{ m/s}^2$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2 \times 0.2}{2.5}} = 0.4 \text{ s}$$

19. (b) For upward motion

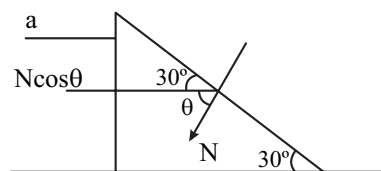
$v = \text{constant}$

$$F = (+2N)$$

For downward motion

$$F = (-2N)$$

20. (c) $N \cos 60^\circ = Ma = 16a \Rightarrow N = 32a$



For block w.r.t wedge $\perp$ to incline

$$N = 8g \cos 30^\circ - 8a \sin 30^\circ$$

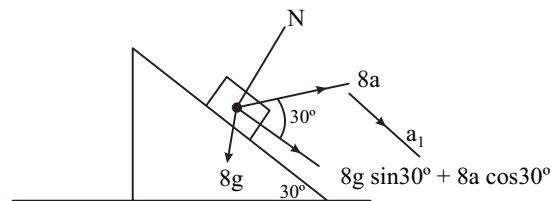
$$\Rightarrow 32a = 8g \cos 30^\circ - 8a \sin 30^\circ$$

$$\Rightarrow 32a = 4\sqrt{3}g - 4a$$

$$\Rightarrow 36a = 4\sqrt{3}g$$

$$\Rightarrow a = \frac{\sqrt{3}}{9}g$$

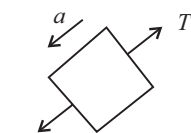
Now for 8 kg, along incline,



$$ma_1 = 8g \sin 30^\circ + 8a \cos 30^\circ$$

$$a_1 = g \times \frac{1}{2} + \frac{\sqrt{3}}{9}g \times \frac{\sqrt{3}}{2} = \frac{g}{2} + \frac{3}{18}g = \frac{2}{3}g$$

21. (b) Free body diagram for the 4 kg block

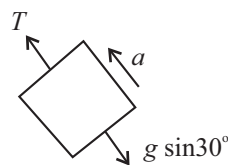


$$4g \sin 60^\circ$$

$$4g \sin 60^\circ - T = 4a$$

...(i)

For the 1 kg block,



$$T - g \sin 30^\circ = a$$

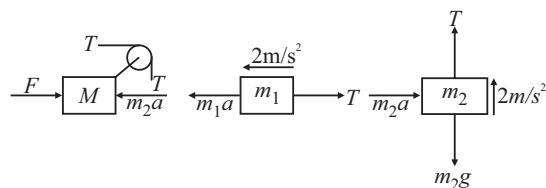
...(ii)

Solving equation (i) and (ii) we get.

$$20\sqrt{3} - T = 4T - 20$$

$$T = 4(\sqrt{3} + 1)N$$

22. (a)



For m_2

$$20 \times 2 = T - 200$$

$$T = 240 \text{ N}$$

For m_1

$$10 \times 2 = 10a - T$$

$$a = 26 \text{ m/s}^2$$

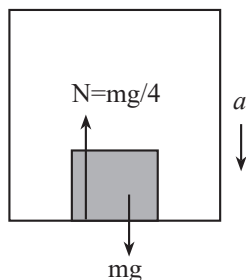
For M

$$Ma = (M + m_2)a + T$$

$$= 120 \times 26 + 240$$

$$= 3360 \text{ N}$$

23. (c) FBD of m

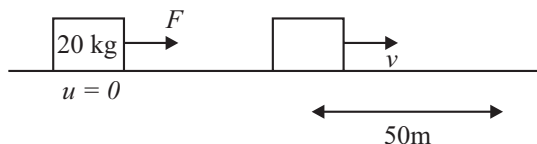


$$W' = m(g - a)$$

$$W' = mg \left(1 - \frac{a}{g} \right) = W \left(1 - \frac{a}{g} \right)$$

$$\frac{W}{4} = W \left[1 - \frac{a}{g} \right] \Rightarrow a = \frac{3g}{4}$$

24. (b)



$$50 = v \times 10$$

$$v = 5 \text{ m/s}$$

$$[\because v = u + at]$$

$$\Rightarrow v = 0 + a \times 20$$

$$a = \frac{1}{4} \text{ m/s}^2$$

$$F = ma = 20 \times \frac{1}{4} = 5 \text{ N}$$

25. [12] $\vec{F} = (2\hat{i} + 3\hat{j} + 5\hat{k}) \text{ N}$ time = 4 sec As body

start from rest therefore position vector initially

$\vec{r}_i = (0\hat{i} + 0\hat{j} + 0\hat{k})$ and u (initial velocity) = 0

given, $\vec{r}_f = (x\hat{i} + y\hat{j} + z\hat{k})$ position vector initially

$\vec{r}_i = (0\hat{i} + 0\hat{j} + 0\hat{k})$ u (initial velocity) = 0 given,

$$\vec{r}_f = (x\hat{i} + y\hat{j} + z\hat{k})$$

Now, from second equation of motion

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \Rightarrow \vec{r}_f - \vec{r}_i = \frac{1}{2} \times \left(\frac{2\hat{i} + 3\hat{j} + 5\hat{k}}{2} \right) \times (4)^2$$

$$\Rightarrow (x\hat{i} + y\hat{j} + z\hat{k}) - (0\hat{i} + 0\hat{j} + 0\hat{k}) = 8\hat{i} + 12\hat{j} + 20\hat{k}$$

$$\Rightarrow x\hat{i} + y\hat{j} + z\hat{k} = 8\hat{i} + 12\hat{j} + 20\hat{k}$$

$\therefore$ The value of $b = 12$

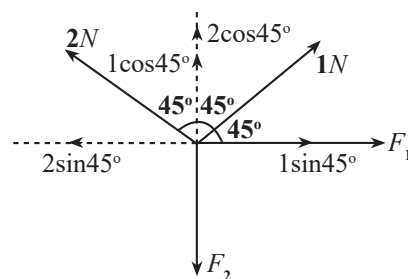
$$26. [3] F_2 = 1\cos 45^\circ + 2\cos 45^\circ = 3\cos 45^\circ = \frac{3}{\sqrt{2}} \text{ N}$$

$$F_1 + 1\sin 45^\circ = 2\sin 45^\circ$$

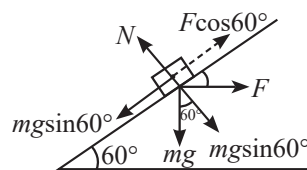
$$F_1 = 1\sin 45^\circ = \frac{1}{\sqrt{2}} \text{ N}$$

$$\frac{F_1}{F_2} = 1:3$$

$$x = 3$$



27. [12] For equilibrium of block along the inclined plane

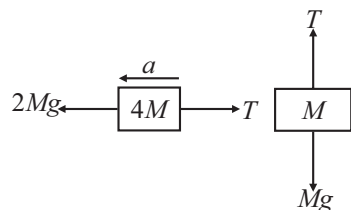


$$F\cos 60^\circ = mg\sin 60^\circ$$

$$F = 0.2 \times 10 \times \frac{\sqrt{3}}{2} \times \frac{2}{1} \Rightarrow F = \sqrt{12} \text{ N}$$

$$\Rightarrow x = 12$$

28. [6]

for M

$$Ma = T - Mg \quad \dots(i)$$

for $4M$

$$4Ma = 2Mg - T \quad \dots(ii)$$

From (i) and (ii)

$$a = \frac{g}{5} \text{ and } T = \frac{6}{5}Mg$$

So $x = 6$

29. [164] Suppose that the magnitude of components along AB and AC are F_1 and F_2 respectively.

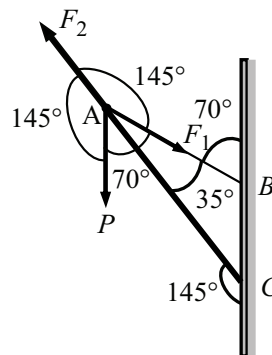
According to the question, $\vec{P} + \vec{F}_1 + \vec{F}_2 = \vec{0}$

By using sign Rule, we can write

$$\frac{F_1}{\sin(145^\circ)} = \frac{F_2}{\sin(70^\circ)} = \frac{P}{\sin(145^\circ)}$$

$$\Rightarrow F_2 = \frac{P \sin(180^\circ - 110^\circ)}{\sin(110^\circ + 35^\circ)}$$

$$\begin{aligned} \Rightarrow F_2 &= \frac{P \sin(110^\circ)}{\sin(110^\circ) \cos(35^\circ) + \cos(110^\circ) \sin(35^\circ)} \\ &= \frac{100 \times 0.939}{0.939 \times 0.819 - 0.342 \times 0.573} \approx 164N \end{aligned}$$



Hence, component of force along AC is $-F_2 = -164N$

30. [492] Net acceleration in vertical direction is a,

$$mg - N = ma$$

$$N = m(g - a) = 60 \times (10 - 1.8) = 60 \times 8.2 = 492N$$

